CODECOMPASS EVALUATION

Hospital-System Bilingual Grounded QA Evaluation

A frozen bilingual QA assessment over one indexed Python repository

READY_WITH_LIMITATIONS

Matrix: hospital_system_bilingual_qa_matrix_v1
Questions: 8
Model: glm-5.3-flash
Status: frozen

Contents

1. Purpose and scope

This report records a frozen bilingual grounded-QA evaluation of CodeCompass over the Hospital-System sample repository. Ground truth was defined before generation and read from source code. The experiment measures retrieval-grounded answer generation, language adherence, and deterministic citation validity for this sample only.

> **Model coverage distinction.** The full eight-question matrix was run with `glm-5.3-flash`. Local Qwen did not run this full matrix. A separate production-path smoke exists in the artifact, but it must not be interpreted as a full-matrix local-vs-cloud comparison.

`HS-QA-01` is a known regression anchor. `HS-QA-02` through `HS-QA-08` are the seven novel frozen questions.

2. Project and index identity

PROJECT ID	SOURCE FILES	SYMBOLS / CHUNKS	VECTORS
2	33	409 / 409	409

Project	Hospital-System	**SQLite/Chroma IDs**	Exact
Embedding	ollama / `nomic-embed-text-local:latest`	**Dimensions**	768
Retrieval	hybrid	**Re-indexed**	False
LLM provider	openai_compatible	**Model**	`glm-5.3-flash`
Temperature	0	**max_tokens**	1024
Executions / retries	8 / 0	**Verdict**	READY_WITH_LIMITATIONS

3. Protocol and scoring rubric

Questions and expected facts were frozen before generation. No tuning occurred during the matrix. Source code was the ground-truth authority, and citation records were checked against SQLite metadata.

PASS	All expected facts are present No invented or forbidden facts Correct target symbol Valid CodeCompass citation/source metadata Answer language matches question language No truncation or provider error
PARTIAL	Stated facts are correct and grounded One or more expected facts are missing No material invented facts
FAIL	Invented or forbidden material fact Contradictory behavior Wrong target symbol Invalid citation/source mapping Provider or truncation failure Materially wrong answer

Matrix SHA-256	`3fbeac128da1251ebc3cd262abcb3208044f058a53e3b94977c5e4fddebbe44d`
Rubric SHA-256	`17cc25b4435b9a44189c2ff87b7e04b800fb1a74a8fbd5de523ef7e53151638e`

4. Aggregate results

PASS	PARTIAL	FAIL	VERDICT
7/8	**0/8**	**1/8**	**READY_WITH_LIMITATIONS**

Slice	Questions	PASS
English	4	3
Persian	4	4
Known regression anchor	1	1
Novel frozen questions	7	6

MEAN LATENCY	MEDIAN LATENCY	MINIMUM	MAXIMUM
8.671s	**8.562s**	**5.494s**	**12.609s**

5. Per-question results

ID	Lang	Category	Target	Result	Latency	Citation
HS-QA-01	EN	attribute_field_enumeration	`Patient.__init__` models/patient.py:6-16	PASS	8.134s	Valid
HS-QA-02	FA	attribute_field_enumeration	`Node.__init__` models/appointment.py:2-5	PASS	5.494s	Valid
HS-QA-03	EN	method_behavior	`Schedule.is_available` models/doctor.py:57-63	PASS	8.990s	Valid
HS-QA-04	FA	method_behavior	`QueueArray.dequeue` data_structures/queue.py:22-33	PASS	7.650s	Valid
HS-QA-05	EN	class_object_relationship	`Doctor.__init__` models/doctor.py:9-23	FAIL	9.910s	Valid
HS-QA-06	FA	class_object_relationship	`Manage.__init__` logic/management.py:7-12	PASS	7.525s	Valid
HS-QA-07	EN	symbol_specific_explanation	`DayManager.next_day` logic/day_manager.py:27-29	PASS	12.609s	Valid
HS-QA-08	FA	symbol_specific_explanation	`File.append` tools/file_handling.py:19-22	PASS	9.059s	Valid

6. Exact captured answers

Answers below are reproduced from the authoritative JSON artifact without semantic rewriting.

HS-QA-01 PASS

8.134s

Question: What attributes are assigned in Patient.__init__?

Target: `models/patient.py`:6-16 / `Patient.__init__`

Expected facts
self.first_name
self.last_name
self.national_id
self.phone_num
self.password
self.gender
self.city
self.full_name
self.insurance_num
self.appointments

In `Patient.__init__` (models/patient.py, lines 6–16), the following attributes are assigned on `self`:

- `first_name`
- `last_name`
- `national_id`
- `phone_num`
- `password`
- `gender`
- `city`
- `full_name` — set to an f-string combining `self.first_name` and `self.last_name`
- `insurance_num`
- `appointments` — initialized to a new `Appointments()` instance

Missing: None
Invented/unsupported: None
Citations: file_id=36, symbol_id=452, chunk_id=42a0c185f2075dcce043fd60f2d6469f3048b722ebc4329967e065691664c5e2, models/patient.py:6-16

Known regression anchor. All 10 expected attributes were present, including self.appointments.

HS-QA-02 PASS

5.494s

Question: سازنده Node در models/appointment.py چه ویژگی‌هایی را مقداردهی می‌کند؟

Target: `models/appointment.py`:2-5 / `Node.__init__`

Expected facts
self.doctor = doctor
self.day = day
self.hour = hour

سازنده `__Node.__init` در فایل `models/appointment.py` سه ویژگی را مقداردهی می‌کند:

- **`self.doctor`** — از پارامتر `doctor`
- **`self.day`** — از پارامتر `day`
- **`self.hour`** — از پارامتر `hour`

کد سازنده (خطوط ۲ تا ۵):

```python
def __init__(self, doctor, day, hour):
    self.doctor = doctor
    self.day = day
    self.hour = hour
```

Missing: None
Invented/unsupported: None
Citations: file_id=34, symbol_id=425, chunk_id=d03f82a50f1788c678260c9cc67483f95577f65bd820f427cdef7d8ec26a0468, models/appointment.py:1-12; file_id=34, symbol_id=426, chunk_id=a94b4ac6724cd2ecf2a0e122d1a9016e512354664f1ebd566a44075575bd2609, models/appointment.py:2-5; file_id=34, symbol_id=428, chunk_id=abaf88ff0dcd4901c9b0a79be40b6b175975a55d7090213cdaba25d9b0308f9c, models/appointment.py:11-12; file_id=34, symbol_id=427, chunk_id=0223f0377d08c35f2cd173d3c4182cebe9d2079553576e40133ff417dcb61add, models/appointment.py:7-9

All expected assignments were present. Additional returned citations were valid but broader than necessary.

HS-QA-03 PASS

8.990s

Question: How does Schedule.is_available decide whether a time slot is available?

Target: `models/doctor.py`:57-63 / `Schedule.is_available`

Expected facts
Converts time to int and subtracts 1 for the list index
Checks self.schedule[day] before reading the slot
Returns True only when the selected slot equals "Available"
Returns False when the day entry is truthy but the selected slot is not "Available"
Returns None implicitly when self.schedule[day] is falsy

Based on the code in `models/doctor.py` (lines 57–63), `Schedule.is_available` works as follows:

1. **Converts the time slot to a 0-based index**: It takes the `time` argument, converts it to an integer, and subtracts 1 (`time = int(time) - 1`). This means callers pass a 1-based slot number, which is mapped to a list index.

2. **Checks that the day exists in the schedule**: It first tests `if self.schedule[day]:` — i.e., the entry for that day must be truthy (non-empty).

3. **Compares the slot's value to the string "Available"**:
   - If `self.schedule[day][time] == "Available"`, it returns `True`.
   - Otherwise, it returns `False`.

One additional behavior visible in the code: if `self.schedule[day]` is falsy (e.g., empty), the outer `if` is not entered and the method has no explicit return in that path, so it implicitly returns `None` — which is falsy and would be treated as "not available" in a boolean context, though it is not an explicit `False`.

Missing: None
Invented/unsupported: None
Citations: file_id=35, symbol_id=438, chunk_id=8164f851e74f01a4f69a42e48ffc67282ab90a9aa4349da9bf235c10ac794e21, models/doctor.py:57-63

All branch-sensitive expected facts were present, including the implicit None path.

HS-QA-04 PASS

7.650s

Question: متد QueueArray.dequeue در صف خالی و غیرخالی چه رفتاری دارد؟

Target: `data_structures/queue.py`:22-33 / `QueueArray.dequeue`

Expected facts
If the queue is empty, returns "Queue is empty."
Otherwise increments self.front by 1
Returns self.queue[self.front - 1] after incrementing front

بر اساس کد موجود در `data_structures/queue.py` (خطوط ۲۲-۳۳)، متد `QueueArray.dequeue` دو رفتار دارد:

صف خالی:
- ابتدا با `self.is_empty()` بررسی می‌کند که صف خالی باشد.
- اگر صف خالی باشد، رشته‌ی `"Queue is empty."` را برمی‌گرداند (یعنی به‌جای پرتاب استثنا، یک پیام متنی برمی‌گرداند).

صف غیرخالی:
- مقدار `self.front` را یک واحد افزایش می‌دهد: `self.front += 1`
- سپس عنصر موجود در اندیس `self.front - 1` (یعنی همان موقعیت قبلیِ `front`) را برمی‌گرداند: `return self.queue[self.front - 1]`

به این ترتیب، عنصر ابتدای صف خوانده شده و نشانگر `front` به سمت جلو حرکت می‌کند.

نکته: در بدنه‌ی متد یک بلوک کد کامنت‌شده نیز وجود دارد که پیاده‌سازی جایگزینی را نشان می‌دهد (خواندن عنصر از `self.queue[self.front]` و سپس کاهش `self.front`)، اما این کد اجرا نمی‌شود و نسخه‌ی فعال همان افزایش `front` است.

Missing: None
Invented/unsupported: None
Citations: file_id=24, symbol_id=316, chunk_id=1df22345e37e9c6c7038efb4a6acb81e56298fbe587c4a4be47acb872f21650b, data_structures/queue.py:22-33

Both active execution paths were explained correctly; the commented alternative was explicitly identified as inactive.

HS-QA-05 FAIL

9.910s

Question: Which helper objects does Doctor.__init__ create and store?

Target: `models/doctor.py`:9-23 / `Doctor.__init__`

Expected facts
self.schedule is initialized with Schedule()
self.appointments is initialized with Appointments()

Based on the provided code, Doctor.__init__ `creates and stores two helper objects:

1. **Schedule()** — a new Schedule `instance assigned to `self.schedule`
2. **Appointments()** — a new Appointments `instance assigned to `self.appointments`

All other attributes set in `__init__`are plain values, not helper objects: the string parameters (first_name,`last_name,`national_id,`medical_num,`phone_num,`address,`city,`speciality,`password), the integer self.visits = 0,`and the f-string self.full_name = f"{self.first_name} {self.last_name}".`

Missing: None
Invented/unsupported: Described unannotated constructor parameters as "string parameters"
Citations: file_id=35, symbol_id=430, chunk_id=9c72cb460e7b3d43f4ccfa12ea1c0cbcea8ff63a35683885deed828231679f3c, models/doctor.py:9-23

The requested helper-object facts were complete, but the unsupported parameter-type claim is a material generation/prose grounding failure under the frozen rubric.

HS-QA-06 PASS

7.525s

Question: سازنده Manage چه مجموعه‌هایی را نگه می‌دارد و چه شیء مدیریت روزی می‌سازد؟

Target: `logic/management.py`:7-12 / `Manage.__init__`

Expected facts
Stores cities in self.cities
Stores specialities in self.specialities
Stores patients in self.patients
Stores doctors in self.doctors
Creates DayManager() and stores it in self.day_manager

بر اساس کد سازندهٔ `Manage.__init__` (فایل `logic/management.py`، خطوط ۷ تا ۱۲):

مجموعه‌هایی که نگه می‌دارد (که همگی به‌عنوان پارامتر ورودی دریافت می‌شوند):
- `self.cities` — شهرها
- `self.specialities` — تخصص‌ها
- `self.patients` — بیماران
- `self.doctors` — پزشکان

شیء مدیریت روزی که می‌سازد:
- `self.day_manager = DayManager()` — یک نمونه از کلاس `DayManager` می‌سازد (این پارامتر ورودی نیست و مستقیماً درون سازنده نمونه‌سازی می‌شود).

Missing: None
Invented/unsupported: None
Citations: file_id=30, symbol_id=381, chunk_id=69c8ba061c1bec5d408854c7261ff2649807ec56b40b998e8a133d9bbf27d2d7, logic/management.py:6-203; file_id=30, symbol_id=382, chunk_id=e5423fb10dff0fafa428a81cff245fca3072c81c2f53b611630c5e234355f617, logic/management.py:7-12; file_id=30, symbol_id=383, chunk_id=4309dea23016534ac81b5feb84e06e0dfdbaa2ac88509b7092a247156e2b3b62, logic/management.py:16-18

All stored dependencies and the constructed DayManager were identified. Additional returned citations were valid but broader than necessary.

HS-QA-07 PASS

12.609s

Question: What exactly does DayManager.next_day do?

Target: `logic/day_manager.py`:27-29 / `DayManager.next_day`

Expected facts
Increments self.curr_day by 1
Calls self.save_day() after incrementing

Based on the provided code, `DayManager.next_day` (defined in `logic/day_manager.py`, lines 27–29) does exactly two things:

1. **Increments the current day**: `self.curr_day += 1` — it adds 1 to the instance attribute `curr_day`.
2. **Saves the day**: it then calls `self.save_day()`.

`python
def next_day(self):
    self.curr_day += 1
    self.save_day()
`

Note: The retrieved context does not include the implementation of `save_day()`, so what that save operation actually does (e.g., where or how the day is persisted) cannot be determined from the provided evidence.

Missing: None
Invented/unsupported: None
Citations: file_id=29, symbol_id=379, chunk_id=c4412c63a3df489c8bac4713e4b7eb9e04dad1b47be79e617a4a923b71bdd817, logic/day_manager.py:27-29

The two operations and their order were correct; the answer appropriately bounded claims about save_day().

HS-QA-08 PASS

9.059s

Question: متد File.append متن ورودی را دقیقاً چگونه به فایل اضافه می‌کند؟

Target: `tools/file_handling.py`:19-22 / `File.append`

Expected facts
Opens self.path in append mode "a"
Writes the supplied line
Writes a newline "\n" immediately afterward

متد `File.append` (در فایل `tools/file_handling.py`، خطوط ۱۹ تا ۲۲) متن ورودی را به این صورت اضافه می‌کند:

1. فایل را با حالت `"a"` (append) باز می‌کند، یعنی محتوا به انتهای فایل اضافه می‌شود و محتوای قبلی حذف نمی‌شود:
```python
with open(self.path, "a") as file:
```
2. متن ورودی (`line`) را با `file.write(line)` می‌نویسد.
3. بلافاصله یک کاراکتر خط جدید (`"
"`) را با `file.write("
")` اضافه می‌کند.

بنابراین متن ورودی دقیقاً همان‌طور که هست (بدون هیچ تغییری) در انتهای فایل نوشته می‌شود و پس از آن یک خط جدید قرار می‌گیرد، به‌طوری که هر فراخوانی بعدی در خط جداگانه‌ای نوشته شود.

Missing: None
Invented/unsupported: None
Citations: file_id=44, symbol_id=518, chunk_id=20fde1b51d208c3d95db9b5108c353ed49af83cd5b6348410fd5b1ca8f01932d, tools/file_handling.py:19-22

The append mode, input write, and trailing newline were all described correctly.

7. Failure analysis

> `HS-QA-05` **- strict FAIL**
> The requested helper-object facts were complete, but the unsupported parameter-type claim is a material generation/prose grounding failure under the frozen rubric.

The requested helper-object facts were correct: `self.schedule = Schedule()` and `self.appointments = Appointments()`. The response also described parameters as string parameters without source evidence. Under the frozen rubric, a material unsupported type claim is FAIL rather than PARTIAL.

This is a generation/prose-grounding limitation. The artifact does not indicate a retrieval failure or citation-mapping failure for this question.

8. Citation and language analysis

CITATION VALIDITY	LANGUAGE ADHERENCE
13/13	**8/8**

All 13 citation records mapped to trusted SQLite metadata. Additional but valid citation noise was observed for `HS-QA-02` and `HS-QA-06`. Citation validity and citation precision are distinct: an extra citation can remain navigationally valid while being less selective than desired.

Persian questions were 4/4 PASS and English questions were 3/4 PASS in this sample. The sample is too small to support a general language comparison.

9. Limitations and verdict

- Small frozen matrix of eight questions from one sample Python repository
- HS-QA-01 is a known regression anchor rather than a novel unseen question
- Only seven questions are novel frozen questions
- One unsupported parameter-type claim was observed
- Additional but valid citation noise was observed for HS-QA-02 and HS-QA-06
- No statistical significance or broad language comparison is supported
- The result does not establish hallucination-free behavior
- Token usage, monetary cost, and provider finish_reason were not retained by the normal Ask API response

READY_WITH_LIMITATIONS

The frozen sample shows strong grounded QA behavior with one unsupported claim. Citation integrity remained valid, but the result does not establish hallucination-free behavior or broad model performance.

10. Reproducibility record

Experiment ID	`hospital_system_bilingual_qa_evaluation_v1`
Matrix ID	`hospital_system_bilingual_qa_matrix_v1`
Matrix SHA-256	`3fbeac128da1251ebc3cd262abcb3208044f058a53e3b94977c5e4fddebbe44d`
Rubric SHA-256	`17cc25b4435b9a44189c2ff87b7e04b800fb1a74a8fbd5de523ef7e53151638e`
Project	2 / Hospital-System
Model	`glm-5.3-flash`
Configuration	hybrid; temperature=0; max_tokens=1024; retries=0
Index	33 files; 409 symbols; 409 chunks; 409 vectors

Authoritative inputs: frozen JSON and CSV artifacts under reports/evaluation. No new generation, retrieval, or benchmark execution was performed for this PDF.